

4.1 If-else branches (general)

PARTICIPATION ACTIVITY

4.1.1: Branching concept.

Start



2x speed

Party of:



Host

If party of 1:

1

Counter

1

Else If party of 2:



2



Else:

4

3



Captions ^

1. A restaurant host seats patrons. The host seats a party of 1 at the counter.
2. A party of 2 is seated at a small table. Larger-sized parties are seated at a large table.
3. The host mentally executes the algorithm: If party of one, seat at the counter; if party of 2, 3, or more, seat at a large table.

branch

A branch is a sequence of statements only executed under a certain condition.

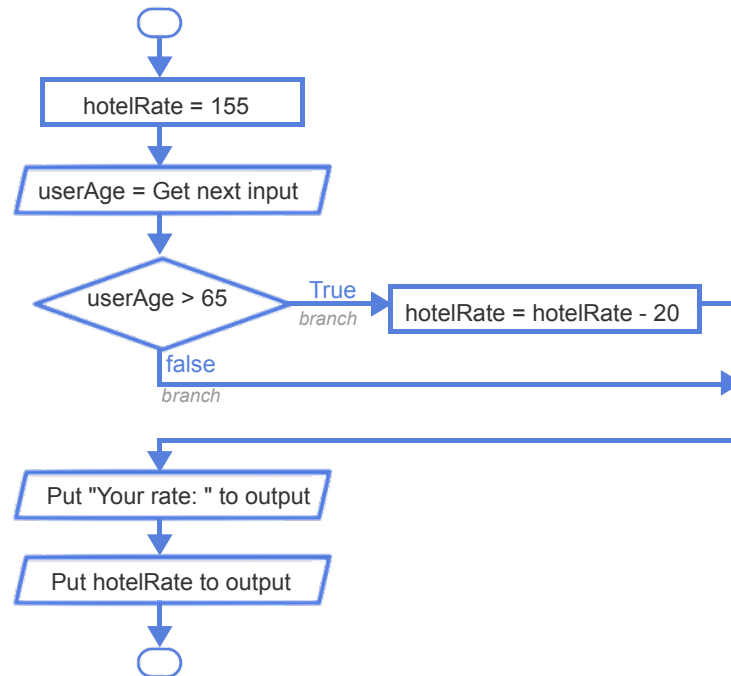
if

An if branch is a branch taken only *if* an expression is true.

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ACTIVITY**

4.1.3: Branches: Hotel rate example.

Start ☐ 2x speed



Variables

135	hotel
68	user

Input

68

Output

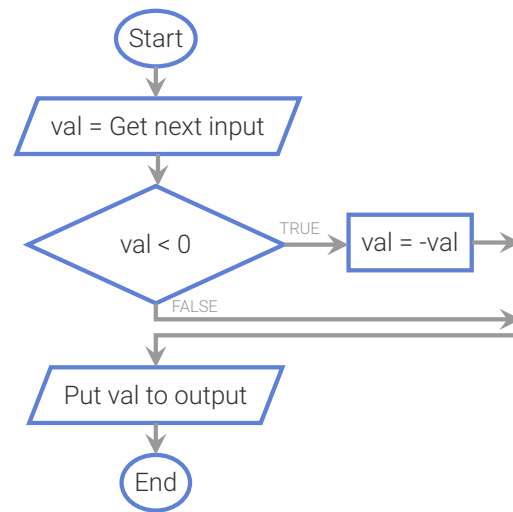
Your rate: 13

Captions ^

1. A decision leads to two program branches. If the expression is True, the first branch executes.
2. If userAge is 68, then $68 > 65$ is true. So the first branch executes, which discounts hotelRate.
3. Execution rejoins the other branch and continues with subsequent statements, outputting "Your rate: 135".

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4.1.5: Computing absolute value.



ENTER EXECUTION

STEP

RUN

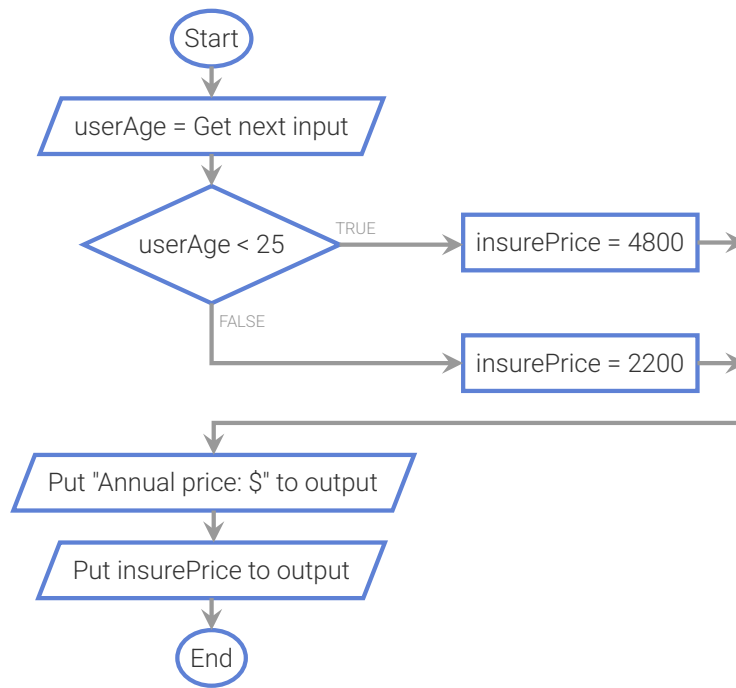
Execution
Medium

if-else

An if-else branch has two branches: The first branch is executed *if* the other branch is executed.

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4.1.7: Insurance price.

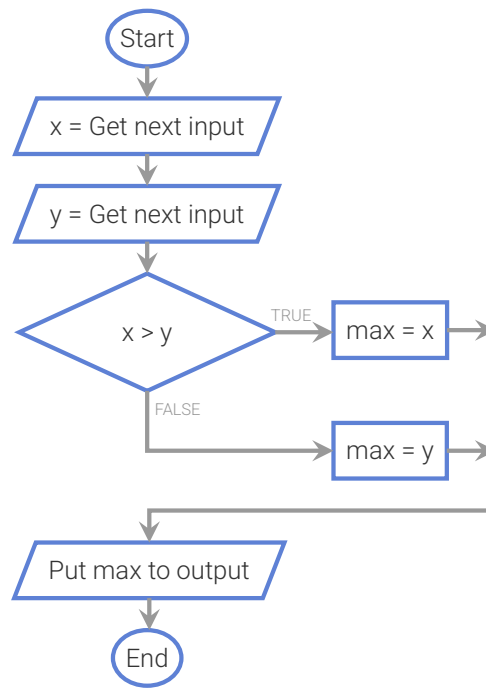
**ENTER EXECUTION**

STEP

RUN

Execution
Medium**PARTICIPATION
ACTIVITY**

4.1.9: If-else branches example: Max.



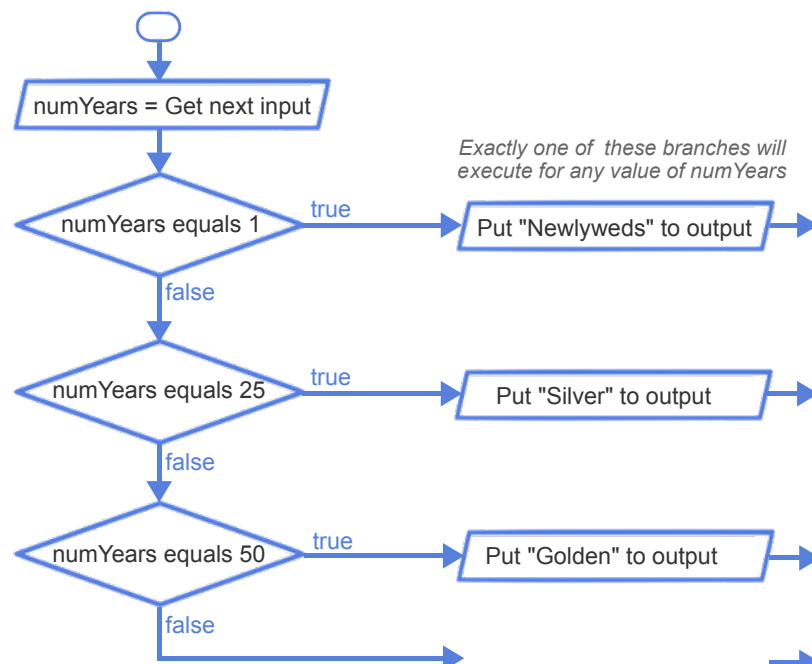
ENTER EXECUTION

STEP

RUN

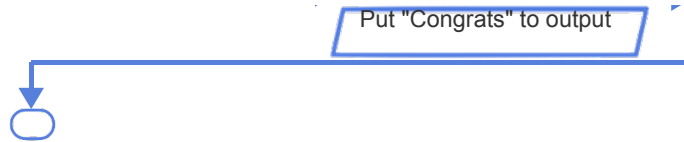
Execution
MediumPARTICIPATION
ACTIVITY

4.1.11: If-elseif example: Anniversaries.

Start ☐ 2x speed

Variables

0 numYears



Captions ^

1. This program detects the specific value of a variable. If numYears is 1, the first branch executes and "Bronze" is output.
2. If numYears is 25, the second branch executes and "Silver" is output. Otherwise, if numYears is 50, the third branch executes and "Golden" is output.
3. Otherwise, the last branch executes.

**CHALLENGE
ACTIVITY**

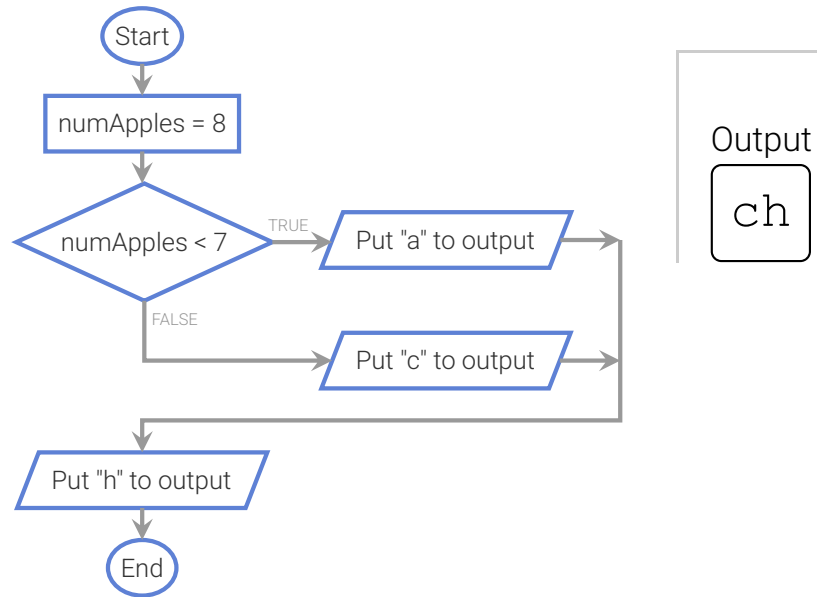
4.1.1: If-else branches.

Note: Level 5 uses `==`, known as the equality operator, to mean equals. Ex: `x == 9` means `x equals 9`.

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Start

Type the program's output



1

2

3

4

Check**Next**View solution ▼ *(Instructors only)*